



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

EMPLOYEE MANAGEMENT SYSTEM

The domain of the Project:

Salesforce Administration & Development

Team Mentors (and their designation):

Sriram Yarrabothula

Salesforce Developer

Submitted By:

Ms. Arpana Sinha

Period of the project

August 2026 to September 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “Employee Management System” has been mentored by Sriram Yarrabothula, organised by SURE Trust, from April 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Submitted By:

Ms. Arpana Sinha

Mentor’s Name: Sriram Yarrabothula

Designation— Salesforce Developer Smartbridge

Prof. Radhakumari

Executive Director & Founder

SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

Project Overview — Employee Management System

Objectives

The main objective of the **Employee Management System** is to develop a centralized Salesforce-based platform for managing employee information and HR operations. The project aims to:

- Maintain employee and department records.
- Manage attendance and leave requests.
- Automate leave approval and email notifications.
- Manage payroll information.
- Provide role-based security for HR, Managers, and Employees.
- Provide dashboards and reports for HR decision-making.

Methods

The system was developed using **Salesforce declarative and programmatic features**, including:

- Custom Objects, Fields, Relationships, and Validation Rules.
- **OWD, Role Hierarchy, and Profiles** for security.
- **Record-Triggered Flows** for leave-status automation and email notifications.
- **Lightning Web Components (LWC)** for employee dashboard, leave requests, and payroll viewing.
- **Reports and Dashboards** for employee, attendance, leave, and payroll analysis.

Key Findings

The completed system demonstrates that Salesforce can effectively centralize HR processes and reduce manual work. Key outcomes include:

- Employee data can be maintained in a structured manner.
- Leave requests can automatically start with **Pending** status.
- Approval processes and notifications can be automated.
- Employees can view relevant leave and payroll information through LWCs.
- Reports and dashboards provide a consolidated view of HR data.
- Role-based access helps protect employee information.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Recommendations

For future enhancement, the system could include:

- Automated payroll calculation and payslip generation.
- More advanced attendance tracking and analytics.
- Mobile-friendly HR functionality.
- Additional approval levels for different leave types.
- Automated HR alerts and reminders.
- Integration with external payroll, biometric attendance, or employee communication systems.
- More advanced analytics using Salesforce reporting and AI capabilities.

Conclusion

The project provides a **centralized, secure, and automated Employee Management System on Salesforce**, integrating employee management, attendance, leave, payroll, security, automation, LWCs, reports, and dashboards into a single platform.

Executive / Stakeholder Highlights

- **Centralized HR Management:** A single Salesforce platform manages **employees, departments, attendance, leave requests, and payroll**, reducing fragmented HR processes.
- **Strong Data Structure:** Related Salesforce objects create a connected HR data model, linking departments with employees and employees with attendance, leave, and payroll records.
- **Improved Data Quality:** Validation rules help prevent incorrect data, including invalid salaries, incorrect leave dates, invalid employee emails, and future attendance entries.
- **Secure Access:** Private OWD, role hierarchy, and separate **HR, Manager, and Employee profiles** provide controlled access to employee information.
- **Process Automation:** Salesforce Flows automate important HR activities such as setting new leave requests to **Pending**, sending approval emails, and identifying missing attendance.
- **Payroll Automation:** Batch Apex and scheduled Apex support **monthly payroll generation and automated payroll execution**.
- **Employee Self-Service:** Lightning Web Components allow employees to view their information, check attendance, submit leave requests, track leave status, and view salary breakdowns.
- **Management Insights:** Reports and dashboards provide visibility into **employees by department, monthly attendance, leave trends, and payroll summaries**, supporting faster HR decision-making.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Project Background, Goals, Scope & Innovation

1. Background and Context of the Project

The **Employee Management System** is a Salesforce-based HR management solution designed to centralize key employee-related activities. The system manages **employee information, departments, attendance, leave requests, and payroll** within one platform. It also uses Salesforce automation, security features, Apex, Lightning Web Components, and reports/dashboards to support HR operations.

2. Problem Statement / Goals

Traditional HR processes can involve separate records and significant manual work for employee, attendance, leave, and payroll management. The goal of this project is to provide a **centralized and automated system** that:

- Maintains accurate employee and department records.
- Tracks employee attendance.
- Manages and monitors leave requests.
- Automates leave status and approval notifications.
- Supports monthly payroll generation.
- Provides secure, role-based access.
- Gives management useful reports and dashboards.

3. Scope and Limitations

Scope:

- Employee and department management.
- Attendance tracking.
- Leave request and approval management.
- Payroll management.
- Role-based security.
- Salesforce Flows and Apex automation.
- Employee self-service through LWCs.
- Reports and dashboards for HR analytics.

Limitations:

- The documented project does not specify integration with external payroll or biometric attendance systems.
- Advanced AI-based HR analytics are not part of the defined implementation.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- The system's functionality is primarily based on the Salesforce platform and its configured objects, automation, and components.

4. Innovation Component

The innovation of the project is the **integration of multiple HR functions into one automated Salesforce ecosystem** rather than managing each activity separately.

Key innovative aspects include:

- **Automated leave management:** New leave requests are automatically assigned **Pending** status and approval can trigger email notification.
- **Automated attendance management:** Scheduled automation can identify employees without attendance records.
- **Automated payroll processing:** Batch Apex and scheduled Apex support monthly payroll generation.
- **Employee self-service:** LWCs provide employees with direct access to their details, attendance, leave status, and payroll information.
- **Real-time management visibility:** Reports and dashboards provide consolidated insights into employee, attendance, leave, and payroll data.

One-line Innovation Statement

“The project innovatively combines Salesforce automation, Apex, security, LWC-based self-service, and analytics to create a centralized and efficient employee management ecosystem.”



Project Objectives

a) Clearly Defined Objectives and Goals of the Project

- **Primary Objective:** Build a comprehensive system to effectively manage employees, departments, attendance, leaves, and payroll.
- **Data Architecture Goals:** Establish a robust data model by creating custom objects and mapping their relationships, such as a Master-Detail relationship for Payroll and 1-to-Many relationships for Departments, Attendance, and Leaves.
- **Data Quality & Security Goals:** Enforce data integrity using Validation Rules and Record Types, while securing the system using a Private Organization-Wide Default (OWD), a defined Role Hierarchy, and custom Profiles.
- **Automation Goals:** Streamline manual processes utilizing Salesforce Flows (Screen, Record-Triggered, and Scheduled) and programmatic Apex (Triggers, Batch, and Schedulable Apex).
- **User Interface & Analytics Goals:** Develop custom user interfaces using Lightning Web Components (LWC) and visualize data through dedicated Reports and Dashboards.

PDF

b) Expected Outcomes and Deliverables

- **Database Deliverables:** A fully configured Salesforce environment containing five custom objects (Employee__c, Department__c, Attendance__c, Leave_Request__c, and Payroll__c) with their corresponding fields and relationships.
- **Declarative Automation Outcomes:** An active Screen Flow for employee creation, a Record-Triggered Flow for leave approvals and email alerts, and a daily Scheduled Flow to mark missing attendance as absent.
- **Programmatic (Apex) Outcomes:** Deployed Apex code including a Trigger to prevent duplicate daily attendance records, and Batch/Schedulable Apex to automatically generate payroll on the 1st of every month.
- **Custom UI Deliverables:** Three functioning Lightning Web Components (LWC): an Employee Dashboard, a Leave Request Component, and a Payroll Viewer.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- **Analytics Deliverables:** Four distinct reporting tools: an "Employees by Department" report, a "Monthly Attendance" report, a "Leave Trends" report, and a centralized "Payroll Summary Dashboard".



Methodology and Results

Methodology and Technology Used

- **Salesforce Administration:** Utilized for core data modeling, establishing object relationships, creating validation rules, and setting up the security architecture.
- **Declarative Automation:** Employed declarative (no-code) methodology using Salesforce Flows to handle standard business logic and user interface routing.
- **Programmatic Development:** Implemented Apex programming for complex backend logic, including Trigger handling for duplicate prevention and Batch/Schedulable classes for recurring processes.
- **Front-End Development:** Utilized the Lightning Web Components (LWC) framework to build dynamic, custom user interfaces for data display and entry.

Tools/Software Used

- **Salesforce Lightning Platform:** The core cloud environment used to build and host the entire application.
- **Salesforce Flow Builder:** The tool used to construct the Screen Flow, Record-Triggered Flow, and Scheduled Flow.
- **Apex:** The object-oriented programming language used to write the Attendance Trigger, Monthly Payroll Batch, and Schedulable classes.
- **Lightning Web Components (LWC):** The UI framework utilized to build the Employee Dashboard, Leave Request Component, and Payroll Viewer.
- **Salesforce Reports & Dashboards:** The native analytics tools used to generate visual summaries of attendance, leaves, and payroll.

Data Collection Approach

- **Interactive User Interfaces:** Employee and department data is collected directly from users via a custom Screen Flow designed for employee onboarding.
- **Self-Service Components:** Leave request data is submitted by users through a custom-built Leave Request Lightning Web Component.
- **Automated System Generation:** Attendance and payroll data points are systematically generated in the background via Scheduled Flows (daily absent marking) and Batch Apex (monthly payroll generation) without manual data entry.

Project Architecture

The system architecture is a multi-layered Salesforce application structured to ensure data integrity, security, and automated processing:

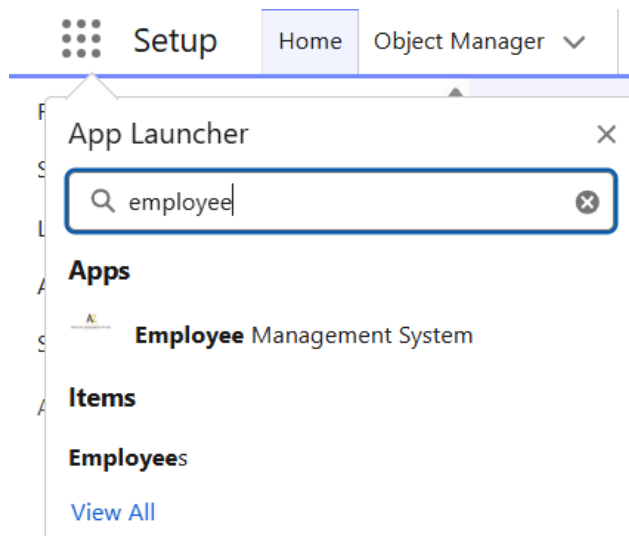
- **Data Layer:** The foundation is built on custom objects (Employee__c, Department__c, Attendance__c, Leave_Request__c, and Payroll__c). The relational architecture relies heavily on Employee__c as the central hub, utilizing a Master-Detail relationship for Payroll and 1-to-Many Lookup relationships for Departments, Attendance, and Leaves. Data quality is maintained at this level via Validation Rules and categorized via Record Types.
- **Security Layer:** Access control is architected with a restrictive foundation using a Private Organization-Wide Default (OWD). Vertical visibility is granted through a strict Role Hierarchy (HR Manager > Team Lead > Employee), and object-level permissions are controlled by custom HR and Manager Profiles.



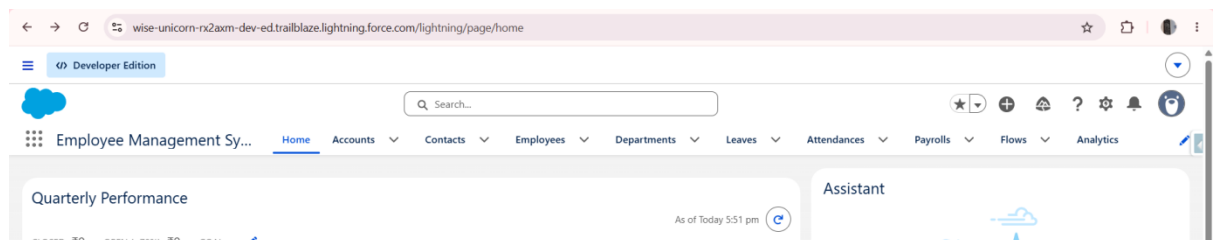
Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

- **Automation Layer:** Business processes are handled by a dual-engine architecture. Standard actions (defaulting statuses, sending emails) are handled declaratively by Flows, while heavy data processing (duplicate prevention, monthly payroll calculation) is delegated to programmatic Apex code.
- **Presentation Layer:** The front-end architecture replaces standard page layouts with custom Lightning Web Components (Employee Dashboard, Leave Request, Payroll Viewer) and centralized visual Dashboards for management reporting

Final project working screenshots along with supporting explanation

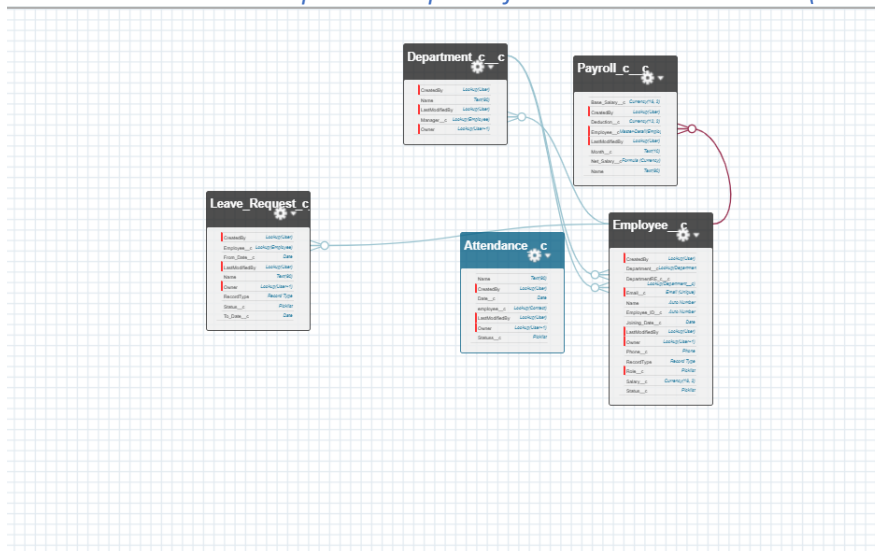


Title: Employee Management System - App Home Page





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



Title: Data Model (Salesforce Schema Builder)

The screenshot shows the Salesforce Flows page for the Employee Management System. The table lists the following flows:

Flow Label	Flow Type	Status	Last Modified Date
1. leaveApproved	Record-Triggered After Save Flow	Activated	08/09/2026, 6:35 pm
2. Leave - Set Pending	Record-Triggered After Save Flow	Activated	08/09/2026, 4:47 pm
3. Employee entry screen	Screen Flow	Activated	08/09/2026, 2:59 pm



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Employee entry screen

Employee Name

Employee Email

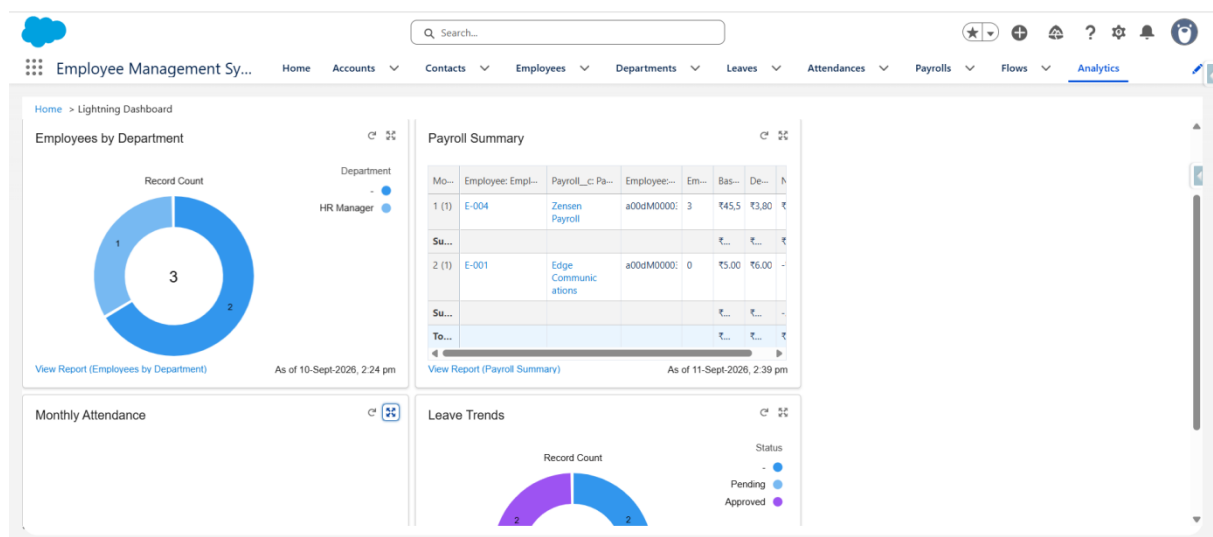
Employee Phone

Salary

Joining Date

[Next](#)

Title: Flow (Automation)



Title: Dash Board & Report

- Project GitHub Link



Overall Experience

Working on the Employee Management System under the mentorship of Sriram Yarrabothula has been a highly transformative professional journey. This project effectively bridged the gap between theoretical Salesforce concepts and hands-on, enterprise-level application development. Translating abstract business requirements into a fully functional, secure cloud application provided a rigorous simulation of industry workflows. One of the most rewarding aspects was troubleshooting complex system faults, such as isolating DML errors during Flow creation, which built immense resilience and confidence in my debugging abilities. Ultimately, this experience solidified my readiness for a professional role in Salesforce development.

New Learnings

In terms of technology, I mastered the use of Salesforce Flow Builder to replace manual data entry, successfully deploying Screen Flows for user interfaces and Record-Triggered Flows for background updates. I also gained hands-on exposure to programmatic development by writing Apex Triggers to enforce complex data validation, alongside designing Batch and Schedulable Apex classes for recurring monthly payroll generation. Furthermore, I learned custom UI architecture by moving beyond standard page layouts and building dynamic Lightning Web Components (LWC) for specialized front-end experiences, such as the Leave Request Component and Employee Dashboard. Finally, I deepened my understanding of relational database design by configuring custom objects and establishing strict Master-Detail and Lookup relationships to maintain data integrity.

On the management side, I learned about data governance and security, specifically how to strategically architect a robust security model to protect sensitive HR data using a Private Organization-Wide Default (OWD) layered with Role Hierarchies and custom Profiles. I developed strong project phasing and execution skills by breaking a massive project down into distinct, manageable phases, including Data Model, Security, Automation, UI, and Analytics. Additionally, I gained the ability to implement business logic by viewing technical features through a business lens, ensuring that every validation rule and automation directly served the company's operational efficiency and compliance requirements.



Conclusion and Future Scope

Recap Objectives and Achievements

The primary objective of this project was to architect a centralized, secure, and automated Employee Management System on the Salesforce platform to streamline human resources operations. This objective was successfully met by developing a robust relational database comprising custom objects for Employees, Departments, Attendance, Leave Requests, and Payroll. Key achievements include the successful enforcement of data integrity through targeted validation rules and the implementation of a strict security model utilizing a Private Organization-Wide Default (OWD) and custom Role Hierarchies. Furthermore, the project successfully replaced manual data entry with declarative automation, deploying Screen Flows for seamless user onboarding, Record-Triggered Flows for dynamic leave approvals, and Scheduled Flows for attendance tracking.

Future Scope of this Project

While the foundational system delivers significant operational efficiency, there is substantial scope for future enhancements. Future development will focus on expanding the user interface by building custom Lightning Web Components (LWC) to create an interactive, self-service Employee Dashboard and advanced Payroll Viewer. Additionally, the system's backend processing capabilities can be advanced by integrating complex Batch Apex classes to automatically handle large-scale payroll deductions and intricate tax calculations. There is also potential to integrate the Salesforce platform with external enterprise resource planning (ERP) software or physical biometric attendance systems to further minimize manual intervention and ensure real-time data accuracy.